

# The genus-1 seven faces: elliptic collision residues and the KZB connection

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## Abstract

At genus 0, the collision residue of a chirally Koszul chiral algebra admits seven equivalent presentations: as a twisting morphism, a line-operator  $r$ -matrix, a  $\lambda$ -bracket, commuting sphere Hamiltonians, the Drinfeld  $r$ -matrix, the Sklyanin bracket, and the Gaudin Hamiltonians. We prove the genus-1 extension: on an elliptic curve  $E_\tau$ , the genus-1 collision residue  $r_{\mathcal{A}}^{(1)}(z, \tau)$  is the elliptic regularization of the genus-0 residue, obtained by replacing rational propagator kernels with prime-form elliptic kernels. All seven faces at genus 1 specialize a *single*  $E_1$ -datum  $\Theta_{\mathcal{A}}$  via the dictionary  $\Psi \rightsquigarrow \hbar \rightsquigarrow \tau$ :  $\Psi$  is the algebraic spectral parameter on genus-0 faces,  $\hbar$  is the quantisation parameter on the Drinfeld / Sklyanin / Yangian faces, and  $\tau$  is the elliptic modulus on the KZB / Belavin–Drinfeld / Felder faces. The passage is the bar-intrinsic projection  $\Theta_{\mathcal{A}}^{(g=1)} = \text{Reg}_\tau \circ \Theta_{\mathcal{A}}^{(g=0)}$  where  $\text{Reg}_\tau$  is elliptic regularisation. For affine Kac–Moody algebras  $\mathcal{A} = \widehat{\mathfrak{g}}_k$ , the seven faces specialize to the Knizhnik–Zamolodchikov–Bernard connection, the Belavin–Drinfeld elliptic  $r$ -matrix, the Felder dynamical elliptic  $R(z, \tau)$  (via Fay trisecant), and the elliptic Gaudin Hamiltonians. The prime-form  $B$ -cycle shift is the local source of the Drinfeld–Kohno quantum-group parameter; with  $\hbar = (k + h^\vee)^{-1}$ , the standard parameter is  $q_{\text{DK}} = \exp(\pi i \hbar)$  and the full-cycle factor is  $q_{\text{DK}}^2$ . For class  $\mathcal{M}$  algebras (Virasoro,  $\mathcal{W}_N$ ), the collision residue produces higher-order elliptic differential operators involving  $P_\tau, P'_\tau, \dots, P_\tau^{(2N-3)}$  outside the affine Hitchin–KZB  $r$ -matrix lane.

*Remark 0.1* (Genus-1 seven-face datum). The genus-1 comparison kernels are obtained by restricting the bar-intrinsic Maurer–Cartan element to the elliptic collision stratum. The KZB, Belavin–Drinfeld, Felder, and Gaudin readouts are different projections of this restricted datum. The prime-form normalization uses

$$\xi_\tau(z) = \partial_z \log \theta_1(z | \tau) = \zeta_\tau(z) - 2\eta_1 z, \quad P_\tau(z) = -\partial_z \xi_\tau(z) = \wp(z, \tau) + 2\eta_1.$$

The quasi-period  $\xi_\tau(z + \tau) - \xi_\tau(z) = -2\pi i$  is the local monodromy input; the quantum-group parameter also requires the chosen Drinfeld–Kohno or Kazhdan–Lusztig normalization.

## I Introduction

The genus-0 seven-face theorem [8] identifies the collision residue  $r_{\mathcal{A}}(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$  of a chirally Koszul chiral algebra  $\mathcal{A}$  with seven classical objects: a bar-cobar twisting morphism, a line-operator  $r$ -matrix in the Dimofte–Niu–Py framework [4], a classical  $\lambda$ -bracket in the Khan–Zeng 3d sigma-model [6], the Gaiotto–Zeng commuting sphere Hamiltonians, the Drinfeld  $r$ -matrix, the Sklyanin Poisson bracket, and the Gaudin Hamiltonians. The seven presentations agree as elements of  $\text{End}_{\mathcal{A}}(2)[[z^{-1}]]$ ; their mutual identification is controlled by the collision depth  $k_{\text{max}}$  and the shadow class  $G/L/C/M$ .

The collision residue  $r_{\mathcal{A}}(z)$  is the degree-2 component of the  $E_1$  Maurer–Cartan element  $\Theta_{\mathcal{A}}^{E_1}$  in the ordered convolution algebra  $\mathfrak{g}_{\mathcal{A}}^{E_1}$ , where  $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$  is the cofree tensor coalgebra with deconcatenation coproduct. Its  $\Sigma_n$ -coinvariant projection is the scalar shadow  $\kappa: \text{av}(r(z))$  in abelian and scalar families, and  $\text{av}(r(z)) + \dim(\mathfrak{g})/2$  for non-abelian affine Kac–Moody; the spectral parameter that  $\kappa$  discards is precisely the data the seven faces read. At genus 1, the spectral parameter acquires modular dependence:  $r_{\mathcal{A}}^{(1)}(z, \tau)$  depends on the elliptic modulus  $\tau$ , and the  $B$ -cycle monodromy of the ordered collision residue encodes the quantum group parameter.

This paper extends the identification to genus 1. The elliptic curve  $E_{\tau} = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$  replaces the Riemann sphere, the rational propagator  $1/z$  is replaced by the prime-form kernel  $\xi_{\tau}(z) = \theta'_1(z, \tau)/\theta_1(z, \tau)$ , and each rational function  $1/z^{n+1}$  in the collision expansion is replaced by the corresponding derivative of  $P_{\tau}(z) = -\partial_z \xi_{\tau}(z)$ . The resulting genus-1 collision residue  $r_{\mathcal{A}}^{(1)}(z, \tau)$  admits seven equivalent presentations that degenerate to the genus-0 faces as  $\tau \rightarrow i\infty$ .

The main results are:

- (1) The *elliptic regularization theorem* (Theorem 2.4): the genus-1 collision residue is

$$r_{\mathcal{A}}^{(1)}(z, \tau) = c_0 \xi_{\tau}(z) + \sum_{n \geq 1} \frac{(-1)^{n-1}}{n!} c_n \cdot P_{\tau}^{(n-1)}(z),$$

where  $c_n = a_{(n)}b$  are the OPE-mode collision coefficients.

- (2) The identification of the  $dz$ -component of the *KZB connection* with the genus-1 collision residue, and of the  $d\tau$ -component with its  $z$ -derivative (Theorem 3.1).
- (3) The identification of the genus-1 collision residue with the *Belavin–Drinfeld elliptic  $r$ -matrix* (Theorem 4.1).
- (4) The *elliptic Gaudin comparison*: the two-point and Cartan sectors match the KZB kernel; full  $n \geq 3$  flatness is the Felder dynamical Yang–Baxter statement (Theorem 5.1).
- (5) The *class  $\mathcal{M}$  collision residue* for Virasoro and  $\mathcal{W}_N$ : higher-order elliptic operators outside the affine Hitchin–KZB  $r$ -matrix lane (Theorems 6.1 and 6.3).
- (6) The *degeneration theorem*: all seven genus-1 faces degenerate continuously to their genus-0 counterparts as  $q = e^{2\pi i \tau} \rightarrow 0$  (Theorem 7.1).

The prime-form kernel defines the elliptic collision residue. Its  $z$ - and  $\tau$ -derivatives give the KZB connection, its root-channel continuation gives the Belavin–Drinfeld kernel, and its collision-depth projections give the Gaudin and class- $\mathcal{M}$  operators.

## 2 The elliptic collision residue

### 2.1 The genus-1 bar propagator

At genus 0 the bar propagator is  $\eta_{12} = d \log(z_1 - z_2)$ : a meromorphic one-form on  $\mathbb{P}^1$  with a simple pole along the diagonal. On the elliptic curve  $E_{\tau}$ , Liouville’s theorem forbids a meromorphic function with a single simple pole that is doubly periodic. The propagator must be replaced by a quasi-periodic object.

**Definition 2.1** (Genus-1 bar propagator). The *genus-1 bar propagator* on  $E_{\tau}$  is the logarithmic derivative of the prime form:

$$\eta_{12}^{E_{\tau}} = d \log E(z_1, z_2) = \xi_{\tau}(z_1 - z_2) d(z_1 - z_2),$$

where  $E(z_1, z_2) = \theta_1(z_1 - z_2 | \tau) / \theta_1'(0 | \tau)$  is the prime form on  $E_\tau$  (a section of  $K^{-1/2} \boxtimes K^{-1/2}$ ),  $\theta_1$  is the Jacobi theta function, and

$$\xi_\tau(z) := \partial_z \log \theta_1(z | \tau) = \frac{\theta_1'(z | \tau)}{\theta_1(z | \tau)} = \zeta_\tau(z) - 2\eta_1 z$$

is the prime-form logarithmic kernel. Here  $\zeta_\tau$  is the Weierstrass zeta function for periods  $(1, \tau)$ :

$$\zeta_\tau(z) = \frac{1}{z} + \sum_{(m,n) \neq (0,0)} \left[ \frac{1}{z - m - n\tau} + \frac{1}{m + n\tau} + \frac{z}{(m + n\tau)^2} \right].$$

Put  $P_\tau(z) := -\partial_z \xi_\tau(z) = \wp(z, \tau) + 2\eta_1$ . The prime-form kernel has Laurent expansion  $\xi_\tau(z) = 1/z + O(z)$  near  $z = 0$ , matching the genus-0 propagator  $1/z$  at leading order. The difference is global:  $\xi_\tau$  is quasi-periodic,

$$\xi_\tau(z + 1) = \xi_\tau(z), \quad \xi_\tau(z + \tau) = \xi_\tau(z) - 2\pi i.$$

The quasi-periodicity of the propagator is the geometric origin of the  $B$ -cycle monodromy that distinguishes genus 1 from genus 0 in all seven faces.

*Remark 2.2* (Propagator weight). The bar propagator  $d \log E(z, w)$  has weight 1 in both variables, regardless of the conformal weight of the fields being sewed. The logarithmic derivative of the prime form is a  $(1, 0)$ -form in each variable; the conformal weight  $h$  of the generator enters the OPE-mode coefficient  $c_n$ , not the propagator kernel.

## 2.2 The collision residue

**Definition 2.3** (Genus-1 collision residue). For a chirally Koszul chiral algebra  $\mathcal{A}$  on  $E_\tau$ , the *genus-1 collision residue* is

$$r_{\mathcal{A}}^{(1)}(z, \tau) := \text{Res}_{1,2}^{\text{coll}}(\Theta_{\mathcal{A}})|_{E_\tau} \in \text{End}_{\mathcal{A}}(2)[[z^{-1}]] \otimes \mathbb{C}[[\tau]],$$

where  $\Theta_{\mathcal{A}}$  is the bar-intrinsic Maurer–Cartan element.

**Theorem 2.4** (Elliptic regularization). *Let  $\mathcal{A}$  be a chirally Koszul chiral algebra with genus-0 collision residue  $r_{\mathcal{A}}(z) = \sum_{n \geq 0} c_n / z^{n+1}$ , where  $c_n = a_{(n)} b$  are the OPE-mode coefficients. The genus-1 collision residue is*

$$r_{\mathcal{A}}^{(1)}(z, \tau) = c_0 \xi_\tau(z) + \sum_{n \geq 1} \frac{(-1)^{n-1}}{n!} c_n \cdot P_\tau^{(n-1)}(z), \quad (2.1)$$

where  $P_\tau^{(m)} = d^m P_\tau / dz^m$ . Since  $P_\tau = \wp + 2\eta_1$ , one has  $P_\tau^{(m)} = \wp^{(m)}$  for  $m \geq 1$ . In particular, the first three terms are

$$r_{\mathcal{A}}^{(1)}(z, \tau) = c_0 \xi_\tau(z) + c_1 P_\tau(z) - \frac{1}{2} c_2 P_\tau'(z) + \dots$$

The Laurent expansion of  $r_{\mathcal{A}}^{(1)}$  near  $z = 0$  agrees with  $r_{\mathcal{A}}(z)$  at leading order.

*Proof.* Each rational function  $1/z^{n+1}$  ( $n \geq 0$ ) in the genus-0 collision expansion is the unique meromorphic function on  $\mathbb{P}^1$  with a pole of order  $n + 1$  at  $z = 0$  and vanishing at  $z = \infty$ . On  $E_\tau$ , the prime-form normalization fixes the corresponding polar kernel. For  $n = 0$  it is

$\xi_\tau(z)$ ; for  $n \geq 1$  it is  $(-1)^{n-1} P_\tau^{(n-1)}(z)/n!$ . The prefactor is fixed by the Laurent expansion  $P_\tau^{(m)}(z) = (-1)^m (m+1)!/z^{m+2} + O(1)$  at  $m = n-1$ :

$$\frac{(-1)^{n-1}}{n!} P_\tau^{(n-1)}(z) = \frac{(-1)^{n-1}}{n!} \cdot \frac{(-1)^{n-1} n!}{z^{n+1}} + O(z^{-n+1}) = \frac{1}{z^{n+1}} + O(z^{-n+1}).$$

Substituting into the propagator-extraction formula of Definition 2.1 yields (2.1).  $\square$

*Remark 2.5* (Quasi-periodicity dichotomy). The kernel  $P_\tau$  and all its derivatives are doubly periodic. The quasi-periodicity of  $r_{\mathcal{A}}^{(1)}(z, \tau)$  is therefore inherited entirely from the  $\xi_\tau$ -term:

$$r_{\mathcal{A}}^{(1)}(z + \tau, \tau) - r_{\mathcal{A}}^{(1)}(z, \tau) = -2\pi i \cdot c_0. \quad (2.2)$$

When  $c_0 = 0$  (the Cartan-type case:  $\beta\gamma$ , lattice VOAs), the collision residue is a genuine meromorphic function on  $E_\tau$ . When  $c_0 \neq 0$  (the Yangian-type case: affine Kac–Moody, Virasoro,  $\mathcal{W}_N$ ), it is quasi-periodic with non-trivial  $B$ -cycle monodromy.

### 3 Face 4: the KZB connection

At genus 0, Face 4 identifies the collision residue with the Gaiotto–Zeng commuting sphere Hamiltonians. At genus 1, the sphere Hamiltonians are replaced by the Knizhnik–Zamolodchikov–Bernard connection on the configuration space  $\text{Conf}_n(E_\tau)$ . The KZB connection has two components: a  $dz$ -component generalizing the genus-0 KZ connection, and a  $d\tau$ -component encoding the dependence of conformal blocks on the modulus  $\tau$ .

The formulas in this section use the KZ-normalized conformal-block kernel. In trace-form normalization the binary affine collision residue is  $k\Omega_{\text{tr}}\xi_\tau(z)$ . The KZ coefficient is  $\Omega_{\text{KZ}}\xi_\tau(z)/(k+h^\vee)$ . These are related by the Sugawara normalization; they are not identical functions of the level.

**Theorem 3.1** (Face 4: KZB connection). *For  $\mathcal{A} = \widehat{\mathfrak{g}}_k$  at level  $k \neq -h^\vee$ , the modular shadow connection on  $\text{Conf}_n(E_\tau)$  restricts to the KZB connection:*

$$\nabla^{\text{KZB}} = d - \underbrace{\frac{1}{k+h^\vee} \sum_{i < j} \Omega_{ij} \xi_\tau(z_{ij}) dz_{ij}}_{dz\text{-component: genus-1 collision residue}} - \underbrace{\frac{1}{k+h^\vee} \sum_{i < j} \Omega_{ij} P_\tau(z_{ij}) d\tau}_{d\tau\text{-component: modular Hamiltonian}},$$

where  $z_{ij} = z_i - z_j$  and  $\Omega_{ij}$  is the Casimir acting on the  $i$ -th and  $j$ -th tensor factors.

(i) *The  $dz$ -component equals the genus-1 collision residue:*

$$H_{z,i}^{\text{KZB}} = \frac{1}{k+h^\vee} \sum_{j \neq i} \Omega_{ij} \xi_\tau(z_{ij}) = \sum_{j \neq i} [r_{\widehat{\mathfrak{g}}_k}^{(1)}(z_{ij}, \tau)]_{ij}. \quad (3.1)$$

(ii) *The  $d\tau$ -component equals the  $z$ -derivative of the collision residue, through the identity  $P_\tau(z) = -\xi'_\tau(z)$ :*

$$H_{\tau,i}^{\text{KZB}} = \frac{1}{k+h^\vee} \sum_{j \neq i} \Omega_{ij} P_\tau(z_{ij}) = - \sum_{j \neq i} \partial_z [r_{\widehat{\mathfrak{g}}_k}^{(1)}(z_{ij}, \tau)]_{ij}. \quad (3.2)$$

*Proof. Part (i).* The affine Kac–Moody OPE  $J^a(z) J^b(w) \sim k \delta^{ab}/(z-w)^2 + f_c^{ab} J^c(w)/(z-w)$  has  $c_0 = \Omega/(k+h^\vee)$  and  $c_n = 0$  for  $n \geq 1$ : the bar propagator absorbs one pole order, so the only

surviving term is  $c_0$ . By Theorem 2.4, the elliptic regularization gives  $r_{\widehat{\mathfrak{g}}_k}^{(1)}(z, \tau) = \Omega \xi_\tau(z)/(k + h^\vee)$  in KZ normalization. Substituting into the Gaiotto–Zeng formula produces (3.1), which is the  $dz$ -component of the KZB connection [3, 7].

**Part (ii).** The  $d\tau$ -component involves the modular Hamiltonian  $D_i$ , whose leading term is  $\sum_{j \neq i} P_\tau(z_{ij}) \Omega_{ij}/(k + h^\vee)$ . The identity  $P_\tau = -\xi'_\tau$  identifies this with  $-\partial_z r_{\mathcal{A}}^{(1)}$ , yielding (3.2).  $\square$

*Remark 3.2* (Shadow connection interpretation). The KZB connection is the genus-1 instance of the modular shadow connection  $\nabla_{g,n}^{\text{hol}}$  on  $\overline{\mathcal{M}}_{g,n}$ : at genus 0 it gives the KZ connection, at genus 1 the KZB connection, and at genus  $\geq 2$  it produces new flat connections on configuration spaces of higher-genus curves.

## 4 Face 5: the elliptic $r$ -matrix

At genus 0, Face 5 identifies the collision residue with the KZ-normalized rational kernel  $\Omega/((k + h^\vee)z)$ . At genus 1, the rational  $r$ -matrix is replaced by the classical elliptic  $r$ -matrix of Belavin [1] and Belavin–Drinfeld [2].

**Theorem 4.1** (Face 5: elliptic  $r$ -matrix). *For  $\mathcal{A} = \widehat{\mathfrak{g}}_k$  with  $\mathfrak{g}$  simple and  $k \neq -h^\vee$ , the KZ-normalized genus-1 two-point kernel equals the classical Belavin elliptic kernel up to the standard level normalization:*

$$r_{\widehat{\mathfrak{g}}_k}^{(1)}(z, \tau) = \frac{1}{k + h^\vee} r_{\mathfrak{g}}^{\text{ell}}(z, \tau). \quad (4.1)$$

For  $\mathfrak{g} = \mathfrak{sl}_2$ , the Belavin  $r$ -matrix is

$$r_{\mathfrak{sl}_2}^{\text{ell}}(z, \tau) = \xi_\tau(z) \cdot \frac{H \otimes H}{2} + \phi_+(z, \tau) E \otimes F + \phi_-(z, \tau) F \otimes E,$$

where  $\{H, E, F\}$  is the standard basis of  $\mathfrak{sl}_2$  and the theta-function ratios are

$$\phi_\pm(z, \tau) = \frac{\theta'_1(0 \mid \tau) \theta_1(z \pm \frac{1}{2} \mid \tau)}{\theta_1(z \mid \tau) \theta_1(\pm \frac{1}{2} \mid \tau)}. \quad (4.2)$$

The full  $n \geq 3$  KZB flatness statement uses the Felder dynamical shift and the dynamical Yang–Baxter equation. It is not a consequence of applying the ordinary rational CYBE to the displayed two-point kernel.

*Proof.* For the Cartan channel: the Kac–Moody OPE with  $c_0 = \Omega/(k + h^\vee)$  gives, after genus-1 regularization,  $\xi_\tau(z) \cdot H \otimes H/2$  in the Cartan direction.

For the root channels: on  $E_\tau$ , the chiral fields  $J^E, J^F$  associated to the positive and negative roots satisfy twisted boundary conditions under  $B$ -cycle monodromy. The collision residue in the root channels acquires the theta-function ratios  $\phi_\pm(z, \tau)$  (4.2), which are the unique meromorphic functions on  $\mathbb{C}$  with a simple pole at  $z = 0$  matching  $1/z$  and with  $B$ -cycle quasi-periodicity dictated by the root lattice [1].

The full  $n$ -point flatness statement belongs to the Felder dynamical KZB lane. The two-point identification (4.1) follows from Theorem 2.4 combined with uniqueness of the classical elliptic  $r$ -matrix [2].  $\square$

*Remark 4.2* (Higher rank). For  $\mathfrak{g} = \mathfrak{sl}_N$  with  $N \geq 3$ , the Belavin elliptic  $r$ -matrix is a matrix-valued meromorphic function on  $E_\tau$  with poles at  $z = 0$  whose residue is the  $\mathfrak{sl}_N$  Casimir. The collision

residue produces this  $r$ -matrix by the same mechanism: elliptic regularization of the Cartan and root channels, with theta-function ratios  $\phi_\alpha(z, \tau)$  indexed by the roots  $\alpha$ . The identification (4.1) holds for all simple  $\mathfrak{g}$ .

## 5 Face 7: the elliptic Gaudin model

At genus 0, Face 7 identifies the collision residue with the Gaudin Hamiltonians  $H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij} / (z_i - z_j)$ . At genus 1, the Gaudin model acquires elliptic structure.

**Theorem 5.1** (Face 7: elliptic Gaudin kernel). *For  $\mathcal{A} = \widehat{\mathfrak{g}}_k$ , the KZ-normalized two-point elliptic Gaudin kernel on  $\text{Conf}_n(E_\tau)$  is*

$$H_i^{\text{ell}} = \frac{1}{k + h^\vee} \sum_{j \neq i} \Omega_{ij} \xi_\tau(z_i - z_j) = \sum_{j \neq i} [r_{\widehat{\mathfrak{g}}_k}^{(1)}(z_i - z_j, \tau)]_{ij}.$$

*After adding the Felder dynamical shift, this kernel is the two-point sector of the flat KZB/Gaudin connection.*

*Proof.* The identification of  $H_i^{\text{ell}}$  with the collision residue is immediate from Theorem 2.4:  $r_{\widehat{\mathfrak{g}}_k}^{(1)}(z, \tau) = \Omega \xi_\tau(z) / (k + h^\vee)$  in KZ normalization. For  $n \geq 3$ , flatness of the elliptic Gaudin connection requires the dynamical Cartan variable and Felder's dynamical Yang–Baxter equation; the non-dynamical two-point kernel displayed here is the affine Cartan/KZB comparison surface.  $\square$

**Remark 5.2** (Hitchin system). The elliptic Gaudin model is the genus-1 instance of the Hitchin integrable system on  $\text{Bun}_G(E_\tau)$ . The Hitchin Hamiltonians are sections of a line bundle on the base of the Hitchin fibration; at genus 1, the base is the affine space of elliptic spectral curves, and the Hitchin Hamiltonians restrict to the elliptic Gaudin Hamiltonians on the trivial  $G$ -bundle. At genus  $g \geq 2$ , the collision residue on  $\Sigma_g$  produces the Hitchin Hamiltonians on  $\text{Bun}_G(\Sigma_g)$ ; this is the structural target of the holographic construction [8].

## 6 Class $\mathcal{M}$ at genus 1: higher-order elliptic operators

For affine Kac–Moody algebras (class  $L$ ), the collision expansion terminates at  $c_0$ : the collision depth is  $k_{\text{max}} = 1$ , and only the  $\xi_\tau$ -term contributes. For class  $\mathcal{M}$  algebras (Virasoro,  $\mathcal{W}_N$ ), the OPE has poles of higher order, and the collision expansion involves  $P_\tau$ ,  $P'_\tau$ , and higher prime-form derivatives.

### 6.1 Virasoro at genus 1

The Virasoro algebra has maximal OPE pole order  $p_{\text{max}} = 4$  (from the  $T(z)T(w)$  OPE). The bar propagator absorbs one order, giving collision depth  $k_{\text{max}} = 3$ . The genus-0 collision residue is

$$r_{\text{Vir}_c}(z) = \frac{c/2}{z^3} + \frac{2T}{z},$$

with  $c_0 = 2T$ ,  $c_1 = 0$  (the  $z^{-2}$  pole is absent from the  $r$ -matrix by a parity argument: the  $d$  log extraction sends  $z^{-2n}$  to  $z^{-(2n-1)}$ ), and  $c_2 = c/2$ .

**Theorem 6.1** (Virasoro genus-1 collision residue). *The genus-1 collision residue of  $\text{Vir}_c$  is*

$$r_{\text{Vir}_c}^{(1)}(z, \tau) = 2T \cdot \xi_\tau(z) - \frac{c}{4} P'_\tau(z). \quad (6.1)$$

*This is an elliptic differential operator of order 2 on conformal blocks on  $E_\tau$ : the  $\xi_\tau$ -term acts by multiplication, while the  $P'_\tau$ -term acts as a second-order differential operator through the stress-energy Ward identity.*

*Proof.* The genus-0 collision expansion has  $c_0 = 2T$ ,  $c_1 = 0$ ,  $c_2 = c/2$ . By Theorem 2.4:

$$r_{\text{Vir}_c}^{(1)}(z, \tau) = c_0 \xi_\tau(z) + c_1 P_\tau(z) - \frac{1}{2} c_2 P'_\tau(z).$$

Since  $c_1 = 0$ , the  $P_\tau$ -term vanishes, and the result is (6.1). The Laurent expansion near  $z = 0$  confirms consistency:  $2T/z - (c/4) \cdot (-2/z^3 + O(z)) = (c/2)/z^3 + 2T/z + O(1)$ , matching  $r_{\text{Vir}_c}(z)$  at leading order.

The  $P'_\tau$ -term acts as a second-order differential operator because the Virasoro OPE at order  $(z - w)^{-4}$  involves  $(c/2) \cdot \text{id}$ , which through the Ward identity produces a second derivative  $\partial_z^2$  on the correlator; the elliptic regularization replaces  $1/z^4$  by  $-P'_\tau(z)/2$ .  $\square$

*Remark 6.2* (The  $c_1 = 0$  vanishing). The vanishing of  $c_1$  is a parity phenomenon: the OPE mode  $T_{(2)}T = \partial T$  is a total derivative, and its contribution to the  $r$ -matrix vanishes after  $d \log$ -extraction. In the elliptic regularization, this means the  $P_\tau(z)$ -term is absent. The first non-trivial correction beyond  $\xi_\tau$  involves  $P'_\tau$ , not  $P_\tau$ .

## 6.2 $\mathcal{W}_N$ at genus 1

The  $\mathcal{W}_N$  algebra has generators  $W_2 = T, W_3, \dots, W_N$  of conformal weights  $2, 3, \dots, N$ . The maximal OPE pole order among generating fields is  $p_{\max} = 2N$  (from the  $W_N$ - $W_N$  OPE), giving collision depth  $k_{\max} = 2N - 1$ .

**Theorem 6.3** ( $\mathcal{W}_N$  genus-1 collision residue). *The genus-1 collision residue of  $\mathcal{W}_N$  involves the prime-form kernel and the derivatives of  $P_\tau$  up to  $P_\tau^{(2N-3)}$ :*

$$r_{\mathcal{W}_N}^{(1)}(z, \tau) = c_0 \xi_\tau(z) + \sum_{\substack{n=1 \\ c_n \neq 0}}^{2N-2} \frac{(-1)^{n-1}}{n!} c_n \cdot P_\tau^{(n-1)}(z), \quad (6.2)$$

*where  $c_n$  are the OPE-mode collision coefficients. The highest-order term is an elliptic differential operator of order  $2N - 2$ . For  $\mathcal{W}_3$ : the collision expansion involves  $\xi_\tau, P_\tau, P'_\tau, P''_\tau, P'''_\tau$ , producing fourth-order elliptic differential operators.*

*Proof.* Direct specialization of Theorem 2.4 to the  $\mathcal{W}_N$  OPE data with  $p_{\max} = 2N$ : the collision depth  $k_{\max} = 2N - 1$  means the genus-0 expansion  $r_{\mathcal{W}_N}(z) = \sum_{n=0}^{2N-2} c_n/z^{n+1}$  has  $2N - 1$  terms, each regularized by  $1/z \mapsto \xi_\tau(z)$  and  $1/z^{n+1} \mapsto (-1)^{n-1} P_\tau^{(n-1)}(z)/n!$  for  $n \geq 1$ .  $\square$

*Remark 6.4* (Outside the affine KZB lane). The elliptic differential operators produced by the class  $\mathcal{M}$  collision residue lie outside the affine Hitchin–KZB  $r$ -matrix lane. The Hitchin system on  $\text{Bun}_G(E_\tau)$  sees the affine Casimir channel. The KZB connection sees the prime-form kernel and its first polar

derivative in the standard affine sector. The class  $\mathcal{M}$  collision residue at depth  $\geq 3$  produces operators involving  $P'_\tau, P''_\tau, \dots, P_\tau^{(2N-3)}$  from higher OPE poles specific to  $\mathcal{W}$ -algebras.

## 7 Degeneration to genus 0

**Theorem 7.1** (Degeneration). *As  $\text{Im}(\tau) \rightarrow \infty$  (equivalently,  $q = e^{2\pi i \tau} \rightarrow 0$ ), all seven genus-1 faces degenerate to their genus-0 counterparts:*

- (i) (Propagator.)  $\xi_\tau(z) \rightarrow \pi \cot(\pi z) \rightarrow 1/z + O(z)$ .
- (ii) (Prime-form derivatives.)  $P_\tau(z) \rightarrow \pi^2/\sin^2(\pi z) \rightarrow 1/z^2 + O(1)$ , and  $P_\tau^{(m)}(z) \rightarrow (-1)^m(m+1)!/z^{m+2} + O(z^{-m})$ .
- (iii) (KZB  $\rightarrow$  KZ.) *The  $dz$ -component degenerates to the KZ connection:  $\xi_\tau(z_{ij}) \rightarrow 1/z_{ij}$ . The  $d\tau$ -component has no genus-0 counterpart.*
- (iv) (Elliptic  $\rightarrow$  rational  $r$ -matrix.) *In trace form,*

$$r_{\text{tr}}^{\text{KM}}(z) = k \Omega_{\text{tr}}/z.$$

*The KZ-normalized connection uses*

$$r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k + h^\vee)z), \quad \Omega_{\text{KZ}} = 2h^\vee \Omega_{\text{tr}}.$$

*The elliptic kernel degenerates through the trigonometric kernel to the corresponding rational kernel in the chosen normalization. These are conventionally related normalizations, not an identity as rational functions of  $k$ .*

- (v) (Elliptic  $\rightarrow$  rational Gaudin.)  $H_i^{\text{ell}} \rightarrow H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$ .

*Proof.* As  $q \rightarrow 0$ , the prime-form kernels have the standard degenerations. The logarithmic derivative satisfies  $\xi_\tau(z) = \pi \cot(\pi z) + O(q)$ , giving  $1/z + O(z)$  for  $|z| < 1$ . Its derivative gives  $P_\tau(z) = \pi^2/\sin^2(\pi z) + O(q)$ , hence  $1/z^2 + O(1)$ . Higher derivatives degenerate analogously.

The KZB degeneration: with  $\xi_\tau(z) \rightarrow 1/z$ , the  $dz$ -component (3.1) becomes  $\sum_{j \neq i} \Omega_{ij}/((k + h^\vee)z_{ij}) dz_{ij}$ , which is the KZ connection.

The elliptic  $r$ -matrix degeneration: as  $q \rightarrow 0$ ,  $\phi_\pm(z, \tau) \rightarrow \pi/\sin(\pi z) + O(q)$ , recovering the trigonometric  $r$ -matrix. A further limit  $z \mapsto z/L$ ,  $L \rightarrow \infty$  gives  $\pi/\sin(\pi z/L) \rightarrow L/z$ , recovering the rational  $r$ -matrix  $\Omega/((k + h^\vee)z)$  in the KZ normalization. Both limits are continuous in the topology of formal Laurent series.  $\square$

The degeneration theorem provides the consistency check between genus-0 and genus-1: every genus-1 face is a continuous deformation of the corresponding genus-0 face, with deformation parameter  $q$ . At  $q = 0$ , the elliptic curve degenerates to the nodal rational curve, and all seven identifications collapse to their genus-0 ancestors.

## 8 Computational verification

The genus-1 seven-face checks are implemented by the compute engine `theorem_genus1_seven_faces_engine.py`, which implements the following tests.



- (1) **Elliptic regularization.** For each standard family  $\mathcal{A}$  in the landscape census, verify that the genus-1 collision residue (2.1) matches the elliptic regularization of the genus-0 collision expansion at  $O(q^0)$  within the stated numerical tolerance.
- (2) **KZB identification.** For  $\mathcal{A} = \widehat{\mathfrak{sl}}_2$  at levels  $k = 1, 2, 3, 5, 10$ , verify that the  $dz$ -component of the KZB connection equals the collision residue, and that the  $d\tau$ -component equals  $-\partial_z r_{\mathcal{A}}^{(1)}$ .
- (3) **Dynamical closure.** Check the two-point and Cartan sectors of the affine elliptic kernel. The full  $n \geq 3$  flatness statement is the Felder dynamical Yang–Baxter closure and is not claimed by this ordinary-kernel test.
- (4) **Gaudin kernel.** Check that the two-point and Cartan elliptic Gaudin kernels agree with the KZB kernel in the chosen normalization. Full  $n \geq 3$  commutativity belongs to the dynamical KZB/Gaudin lane.
- (5) **Degeneration.** Verify that  $\|r_{\mathcal{A}}^{(1)}(z, \tau) - r_{\mathcal{A}}(z)\|/\|r_{\mathcal{A}}(z)\| < 10^{-6}$  for  $\text{Im}(\tau) \geq 12$  across all standard families.
- (6) **Class  $M$  structure.** For  $\text{Vir}_c$  at  $c = 1, 1/2, 25, 26$ , verify (6.1) and confirm the  $c_1 = 0$  vanishing. For  $\mathcal{W}_3$  at  $c = 2, 50, 98$ , verify (6.2) and confirm terms through  $P_{\tau}'''$ .

## 9 The Verlinde formula from ordered chiral homology

At integer level  $k \in \mathbb{Z}_{\geq 1}$ , the ordered chiral homology of  $\widehat{\mathfrak{g}}_k$  on  $E_{\tau}$  recovers the Verlinde partition function. The connection to the genus-1 seven-face identification is direct: the KZB connection of Theorem 3.1 has finite monodromy at integer level, and the trace of the  $B$ -cycle monodromy over the space of conformal blocks recovers the Verlinde formula.

**Proposition 9.1** (Verlinde from genus-1 collision residue). *For  $\mathcal{A} = \widehat{\mathfrak{g}}_k$  at integer level  $k \geq 1$ , the  $B$ -cycle shift of the KZ-normalized prime-form kernel is*

$$\mathcal{M}_B^{\text{ker}} = \exp\left(-\frac{2\pi i}{k + h^{\vee}} \Omega\right) = \exp(-2\pi i c_0),$$

where  $c_0 = \Omega/(k + h^{\vee})$  is the degree-0 collision coefficient. The quantum group parameter is  $q_{\text{DK}} = \exp(\pi i/(k + h^{\vee}))$ , with full-cycle factor  $q_{\text{DK}}^2 = \exp(2\pi i/(k + h^{\vee}))$ . The Verlinde partition function at genus 1 is

$$Z_1(\widehat{\mathfrak{g}}_k) = \sum_{\lambda \in P_+^k} |S_{0\lambda}|^0 = |P_+^k|,$$

where  $P_+^k$  is the set of integrable highest weights at level  $k$  and  $S_{0\lambda}$  are the modular  $S$ -matrix entries. At higher genus  $g$ , the Verlinde formula  $Z_g = \sum_{\lambda} S_{0\lambda}^{2-2g}$  is recovered from the ordered chiral homology via handle attachment and separating factorization of the ordered bar complex.

**Remark 9.2** (The  $B$ -cycle monodromy and the quantum group). The quasi-periodicity (2.2) of the genus-1 collision residue is the geometric origin of the quantum group parameter. For affine  $\widehat{\mathfrak{sl}}_{2,k}$ : the  $B$ -cycle monodromy eigenvalues are written in the Drinfeld–Kohno normalization with  $q_{\text{DK}} = e^{\pi i/(k+2)}$ , and the full cycle contributes  $q_{\text{DK}}^2$ . The passage from the  $E_i$  collision residue  $r^{(1)}(z, \tau)$  to the quantum group  $U_{q_{\text{DK}}}(\mathfrak{g})$  is the  $B$ -cycle monodromy of the ordered bar complex: the spectral parameter  $z$  (genus-0 data) becomes the quantum group parameter through exponentiation of the quasi-period.

## 10 The critical level

The genus-1 seven-face identification has a distinguished boundary: the critical level  $k = -h^\vee$ , where the Sugawara construction degenerates. The collision residue and the KZB connection undergo a qualitative change at this value.

**Theorem 10.1** (Critical level jump at genus 1). *For  $\mathcal{A} = \widehat{\mathfrak{g}}_k$  at the critical level  $k = -h^\vee$ :*

- (i) *The shadow invariant vanishes:  $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$ . The KZB-normalized kernel is singular at this level, so the non-critical monodromy formula has no regular value there.*
- (ii) *The degree-2 ordered chiral homology on  $V \otimes V$  doubles:  $\dim H^1_{\text{ord},2}$  jumps from 4 (generic level) to 8 (critical level) for  $\mathfrak{g} = \mathfrak{sl}_2$  (finite computation,  $\chi_{\text{ord},2} = -4$  conserved). This is distinct from the full symmetric bar cohomology at critical level, addressed in (iii).*
- (iii) *Under the completed critical bar–Ext comparison, the critical bar–Ext groups identify with differential forms on the space of  $\mathfrak{g}^\vee$ -opers:*

$$H^n_{\text{bExt}}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \Omega^n(\text{Op}_{\mathfrak{g}^\vee}(D)).$$

*This is a completed critical statement, not an equality for the raw symmetric bar cohomology.*

- (iv) *The genus-1 collision residue degenerates: the KZ normalization  $r^{(1)}(z, \tau) = \Omega \xi_\tau(z)/(k + h^\vee)$  has a pole at  $k = -h^\vee$ , and the KZB connection becomes singular. The critical level requires a separate regularization via the Feigin–Frenkel center [5].*

**Remark 10.2** (Koszulness at critical level). At generic level  $k \neq -h^\vee$ , the affine Kac–Moody algebra  $\widehat{\mathfrak{g}}_k$  is chirally Koszul, and the seven-face identification applies without modification. At the critical level  $k = -h^\vee$ , the KZ/KZB normalization and Sugawara topologisation fail, and the Feigin–Frenkel centre appears. This does not by itself negate PBW Koszulness of the universal affine algebra; it moves the statement to the separate critical-level bar-cohomology lane. The seven KZ faces degenerate: the Gaudin Hamiltonians become the Feigin–Frenkel higher Gaudin Hamiltonians, and the  $r$ -matrix limit requires the Langlands dual group  ${}^L G$  via the geometric Langlands correspondence. The critical level is a boundary of the non-critical seven-face identification.

**Remark 10.3** (Spectral sequence at critical level). The passage from generic to critical level is controlled by a spectral sequence with  $d_1 = \lambda \cdot [\partial]$ , where  $\lambda = k + h^\vee$  is the level shift and  $\partial$  is the boundary class. At  $\lambda = 0$  (critical level),  $d_1$  vanishes, and the spectral sequence degenerates at  $E_1$ : the bar cohomology jumps because the differential that would kill the extra classes is absent. This is the mechanism by which the degree-2 ordered chiral homology  $H^1_{\text{ord},2}$  doubles from 4 to 8 for  $\mathfrak{sl}_2$ . For the completed critical bar–Ext cohomology at critical level, the  $d_1$ -vanishing lets the entire de Rham algebra  $\Omega^*(\text{Op}_{\mathfrak{sl}_2}(D))$  appear in completed critical bar–Ext cohomology, which is infinite-dimensional.

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